

使用CompletableFuture之四



如何处理异步代码中的异常

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处理异步代码中的异常-1

ExceptionDemos.java

```
private static void dealWithException() {  
    var future : CompletableFuture<Object> = CompletableFuture.supplyAsync(() -> {  
        throw new IllegalStateException("故意抛出的异常");  
        //return 100;  
    });  
    try {  
        System.out.println(future.get());  
    } catch (ExecutionException | InterruptedException e) {  
        System.out.println("捕获到异常一" + e.getMessage());  
    }  
}
```



要在get()方法调用的地方那儿进行捕获。

处理异步代码中的异常-2

可以使用`exceptionally()`“吃掉”异常，它会返回一个新的`CompletableFuture`对象。

```
private static void dealWithException2() {  
    var future : CompletableFuture<Object> = CompletableFuture.supplyAsync(() -> {  
        throw new IllegalStateException("故意抛出的异常");  
    });  
    try {  
        var result : Object = future.exceptionally(e -> {  
            System.out.println("exceptionally()捕获到异常—" + e.getMessage());  
            return 100;  
        }).get();  
        System.out.println("结果为: " + result);  
    } catch (ExecutionException | InterruptedException e) {  
        System.out.println("捕获到异常—" + e.getMessage());  
    }  
}
```

显式设置异常

```
1 usage
private static void explicitSetException(){
    CompletableFuture<String> future=new CompletableFuture<>();
    new Thread()->{
        future.completeExceptionally(new Exception("显式设置异常"));
    }.start();
    try{
        future.join();
    }catch (Exception e){
        System.out.println(e.getMessage());
    }
}
```

Run: ExceptionDemos x

▶ ↑ "C:\Program Files\Java\jdk-17.0.3.1\bin\java.exe"
⚙ ↓ java.lang.Exception: 显式设置异常
■ ↻ Process finished with exit code 0